Task.1

Airfoil Selection for Two RC Aircraft

Technical report. Plane A: maximum stable payload. Plane B: maneuverability.

1. Aim and assumptions

This report selects one airfoil for each of two radio-controlled aircraft.

Plane A must carry as much payload as possible while flying stably.

Plane B must turn, roll and loop well.

Assumptions:

• Small RC aircraft flying at low speed, so Reynolds number is roughly 200,000 (chord about 0.2 m, speed about 15 m/s).

• Airfoils are compared as 2D sections; wing area, weight and motor are treated as separate design choices.

Shortlisted foils

For Maximum stable payload plane a rc with design like clark Y and selig S1223 are better are they are thicker and have high chamber

For the other type the airfoil can be designed symmetrically like in Naca 0012

Comparison

In clark Y and selig s1223 on seeing results clark Y come out with greater stability as compare to s1223 as it need higher drag and a stronger tail

which makes a heavily loaded air craft harder to stable

While in clark Y and Naca 0012

Due to it's design naca 0012 can obeys it's commands fastly compared to clark

Clark y generates lift itself at a low speeds because of its airfoil which cannot be done by naca0012

Selection and justification

Plane A: Clark Y

Plane A needs high lift at low speed and a forgiving stall. The Clark Y is 11.7% thick, flat on its lower surface aft of 30% chord, Its camber produces lift at zero angle of attack, which helps carry weight. The flat underside also makes the wing easy to build and trim. The S1223 gives more lift but with higher drag and a strong nose-down pitching moment, which makes a heavily loaded aircraft harder to keep stable.

Plane B: NACA 0012

Plane B must behave the same way upright and inverted. A symmetric airfoil has zero lift at zero angle of attack, so lift is controlled mainly by tilt and the wing flies equally well inverted . At 12% thickness it is strong enough for aerobatic loads, has low drag, and stalls predictably.

How airfoil geometry affects performance

• Camber raises lift at a given angle and moves the zero-lift angle below zero. More camber gives more payload capacity but more nose-down

pitching moment and worse inverted flight.

• Thickness gives structural depth and softens the stall, but increases drag.

• Leading-edge radius controls how smoothly the flow stays attached at high angles; a rounder nose gives a gentler stall.

• Lift-to-drag ratio sets how efficiently lift is produced, which matters for payload flight time and motor load.

Can one airfoil satisfy both missions?

Only as a compromise. Plane A wants camber for lift and Plane B wants symmetry for inverted flight and neutral response. A lightly cambered section such as the NACA 2412 carries moderate payload and tolerates inverted flight, but it is worse than the specialist airfoils at each mission. A symmetric wing can still carry payload if the wing area is increased, but it then needs more speed or a larger wing, which adds weight and reduces agility. A highly cambered wing can be flown inverted, but it needs a large control input to hold level inverted flight, which hurts precision aerobatics.

Airfoil versus whole-aircraft factors

The airfoil sets lift and drag per unit of wing. Other factors set what the aircraft actually does:

• Wing loading (weight divided by wing area) determines stall speed and payload capacity [3].

• Control surfaces and tail design determine roll rate, pitch response and trim.

• Structure decides whether the aircraft survives payload loads or aerobatic g-loads.

• Thrust and power-to-weight decide climb, vertical maneuvers and takeoff with payload.

A larger wing or stronger motor can offset a weaker airfoil, so the airfoil alone does not decide performance.

8. Conclusion

Clark Y is selected for Plane A for its balance of lift, low drag and gentle stall. NACA 0012 is selected for Plane B for symmetric, predictable behavior in all attitudes. One airfoil cannot fully serve both missions without a performance compromise.

References

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